

**M Sc (BIOTECHNOLOGY/ MICROBIOLOGY/ BIOCHEMISTRY)**  
**SEMESTER I EXAMINATION (BACKLOG)**  
**BT703/MI703/BC703: - BIOINSTRUMENTATION AND BIOSTATISTICS**

**Date: 28.04.2011**

**Time: 10:00 AM to 1:00 PM**

**Total Marks: 70**

**Instructions to the candidates:**

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

**PART A**

**Total marks: 20**

**Q. 1 Choose the correct option and put  $\sqrt$  mark in front of it:**

- 1 PET stands for  
(a) Positive emission tomography (b) Positron emission tomography  
(c) Proton emission tomography (d) All the above
- 2 Nozzle is an integral part of  
(a) Thermocycler (b) Flowcytometer  
(c) Mass Spectroscope (d) Density Gradient Centrifuge
- 3 A lens with a short focal length will magnify an object more than a weaker lens having a longer focal length  
☐ True ☐ False
- 4 Theta in numerical aperture is defined as half the angle of the cone of light entering an objective  
☐ True ☐ False
- 5 Linear Polarizer is used in  
(a) Circular dichroism spectroscopy (b) Mass Spectroscopy  
(c) UV-Visible Spectroscopy (d) Fluorescence Spectroscopy
- 6 RCF stands for  
(a) Relative centrifugal friction (b) Relative centrifugal force  
(c) Relative centripetal field (d) Relative centrifugal friction
- 7 Immunoprecipitation can be detected in 1% agarose gel  
☐ True ☐ False
- 8 Svedberg unit is associate with  
(a) Ultra centrifugation (b) Fluorescence activated cell sorter  
(c) Infrared and Raman Spectroscopy (d) Patch clam recording
- 9 HPLC stands for  
(a) High performance liquid chromatography (b) High profile liquid chromatography  
(c) High point liquid chromatography (d) None of the above



- 10 Agarose gels can be used for the electrophoresis of proteins as well as nucleic acids  
☐ True ☐ False
- 11 Radiodating is done for  
(a) Age determination (b) Mass determination  
(c) Charge determination (d) None of the above
- 12 DNA is mostly stained with ethidium bromide in agarose gels for visualization  
☐ True ☐ False
- 13 Thermocycler is used for  
(a) PCR (b) Western blotting  
(c) SDS-PAGE (d) All the above
- 14 RAPD stands for randomly amplified polyclonal DNA  
☐ True ☐ False
- 15 Second dimension in two dimensional electrophoresis  
(a) IEF (b) SDS-PAGE  
(c) Northern blot (d) All the above
- 16 \_\_\_\_\_ method is used for gene analysis at RNA level  
(a) Western blotting (b) Northern blotting  
(c) Southern blotting (d) None of the above
- 17 The survival time in plasma of incompletely desialylated ceruloplasmin, labeled with  $^{64}\text{Cu}$  or with tritium, was studied in the rat. Removal of less than 20% of the terminal sialic acid, with consequent exposure of the penultimate galactosyl residues, resulted in the rapid disappearance from the circulation of more than 50% of the injected dose. The number of sialyl residues removed from a given molecule is  
(a) A Nominal Variable (b) An Ordinal Variable  
(c) A Continuous Variable (d) A Discrete Variable
- 18 Knowledge about the sizes of peptides generated by proteasomes during protein degradation is essential to fully understand their degradative mechanisms and the subsequent steps in protein turnover and generation of major histocompatibility complex class I antigenic peptides. The Size of peptides is a continuous variable  
☐ True ☐ False
- 19 A 0.1 M solution of acid was used to titrate 10 ml of 0.1 M solution of alkali and the following volumes (ml) of acid were recorded: 9.88; 10.18; 10.23; 10.39; 10.21  
The Mean Volume of acid is \_\_\_\_\_
- 20 A 0.1 M solution of acid was used to titrate 10 ml of 0.1 M solution of alkali and the following volumes (ml) of acid were recorded: 9.88; 10.18; 10.23; 10.39; 10.21  
The Median Volume of acid is \_\_\_\_\_



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**Total Marks: 50**

**Part B**

**Section I**

**Q.1 A. Write a note on the following (any two)** (5)

- Light Microscope
- Dark-Field Microscope
- Atomic Force Microscopy
- Transmission Electron Microscope

**Q.1 B. i. Diagrammatically describe working of a flowcytometer?** (3)  
**ii. Write in short the procedure of double diffusion immunoprecipitation method** (2)

**Or**

**i. Describe nick translation method for probe preparation in FISH?** (3)  
**ii. Write a short note on sandwich ELISA** (2)

**Q.2 A. Write a note on any one of the following** (4)

- (i) Types of Centrifugation
- (ii) Spectrofluorimetry

**Q.2 B. Describe briefly (any two)** (6)

- (i) ECG
- (ii) MS
- (iii) CD spectroscopy
- (iv) UV/ Visible spectroscopy

**Section II**

**Q.3 (A) Write note on any one** (5)

- (i) SDS-PAGE
- (ii) DNA isolation

**Q.3 (B) Briefly describe any one** (5)

- (i) PCR and its applications
- (ii) Western blotting



**Q.4 (A) Write a note on any two**

(5)

- (i) HPLC
- (ii) Types of radioactive decay
- (iii) Agarose gel electrophoresis of human DNA
- (iv) Capillary electrophoresis

**Q.4 (B) Write a note on any two**

(5)

- (i) Capillary Chromatography
- (ii) Applications of radioisotopes in biological sciences
- (iii) Autoradiography
- (iv) Gradient gels

**Q.5 Attempt any two**

(10)

- (i) A pipette was used to transfer  $1 \text{ cm}^3$  of distilled water into a tared weighing boat. The weight of water delivered was determined on an analytical balance. The experiment was performed 50 times and gave the following results (g water)

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.948 | 1.012 | 1.085 | 1.063 | 1.01  | 1     | 0.994 | 0.986 | 0.995 | 0.999 |
| 0.977 | 0.957 | 0.946 | 0.96  | 0.955 | 1.01  | 0.965 | 0.975 | 0.972 | 0.957 |
| 0.989 | 0.974 | 0.98  | 0.98  | 1.001 | 0.977 | 1.021 | 1.051 | 0.965 | 0.963 |
| 0.984 | 0.978 | 0.968 | 0.96  | 1.027 | 0.959 | 0.985 | 0.985 | 0.967 | 0.96  |
| 0.969 | 0.965 | 0.945 | 0.961 | 0.975 | 0.988 | 0.971 | 0.983 | 0.962 | 0.992 |

Using class width 0.02, construct a frequency distribution of weight.  
The first limit is 0.940 – 0.959

- (ii) One hundred microscope fields on a single slide were examined and the number of bacteria counted per field was determined. The data are presented in frequency table below. Compute Mean and Standard Deviation of number of bacteria per field

| Number of bacteria per field (x) | Number of fields | Number of bacteria per field (x) | Number of fields |
|----------------------------------|------------------|----------------------------------|------------------|
| 4                                | 2                | 16                               | 4                |
| 5                                | 4                | 17                               | 7                |
| 6                                | 3                | 18                               | 7                |
| 7                                | 8                | 19                               | 4                |
| 8                                | 5                | 20                               | 1                |
| 9                                | 7                | 21                               | 2                |
| 10                               | 11               | 22                               | 0                |
| 11                               | 7                | 23                               | 3                |
| 12                               | 4                | 24                               | 1                |
| 13                               | 8                | 25                               | 0                |
| 14                               | 6                | 26                               | 1                |
| 15                               | 5                | 27                               | 0                |
| 16                               | 4                |                                  |                  |



- (iii) In pharmacologic research a variety of clinical chemistry measurements are routinely monitored closely for evidence of side effects of the medication under study. Suppose typical blood – glucose levels are normally distributed with mean 90 mg/dL and standard deviation 38 mg/dL. If the normal range is 65 – 120 mg/dL, what percentage of values will fall in the normal range?
- (iv) One of the most common laboratory tests performed on any routine medical examination is a blood count. The two main aspects to a blood count are (1) counting the number of white blood cells (“white count”) and (2) differentiating white blood cells that do exist into five categories – namely, neutrophils, lymphocytes, monocytes, eosinophils, and basophils. Both the white count and the differentials are extensively used in making clinical diagnoses. If the distribution of the number of neutrophils out of 5 blood cells taken at random has binomial distribution, given the probability that any one cell is a neutrophils is 0.6, what is the probability that any 2 cells out of 5 will be neutrophils?
- (v) Three analysts each made six determinations of the paracetamol content of the same batch of tablets. The results are shown below:

| Analyst | Paracetamol content (% m/m) |       |       |       |       |       |
|---------|-----------------------------|-------|-------|-------|-------|-------|
| A       | 84.32                       | 84.51 | 84.63 | 84.61 | 84.64 | 84.51 |
| B       | 84.24                       | 84.25 | 84.41 | 84.13 | 84.0  | 84.3  |
| C       | 84.29                       | 84.4  | 84.68 | 84.28 | 84.4  | 84.36 |

Test whether there is any significant difference between the means obtained by the three analysts. Use 5% level of significance.